VLN Tutorial

Chapter2 Benchmarks and Simulators

Conner QIU

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01

R2R Introduction

R2R benchmark

Walk toward the railing and right past the stairs. Walk into the living room and turn right. Stop by the end table.

A navigation benchmark = scene dataset + simulator + task dataset

- R2R is based on the MP3D scene dataset,
- It uses the MP3D simulator,
- The navigation space is a pre-defined graph.
- It provides annotated instructions and path

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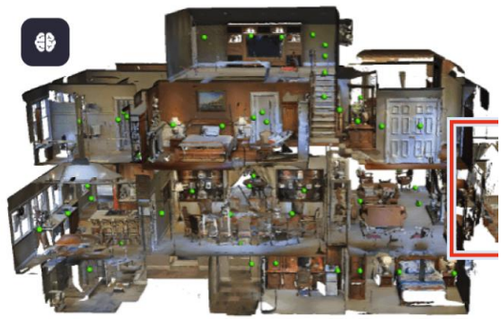
Scene Dataset

Type of scene datasets

- Scan and then reconstruction: MP3D, HM3D
- Synthetic scene datasets: HSSD, ProcTHOR

Think： Pros and cons of reconstruction and synthetic datasets?

Visualization：



MP3D

HM3D

HSSD

ProcTHOR

Links:
MP3D: https://niessner.github.io/Matterport/
HM3D: https://aihabitat.org/datasets/hm3d/
HSSD: https://3dlg-hcvc.github.io/hssd/
ProcTHOR: https://procthor.allenai.org/

MP3D – The one R2R uses

Information contatined：

- panoramic views – 360 degree view
  - 12 * 3 RBG-D images
- RGB-D images
- Does not contain category label

	Total				Average per scene				
Dataset	Scenes	Objects	Categories	Nav area	Nav area	Nav comp	Clutter	Categories	Instances
HM3DSem (scans) [43]	181	59,269	1,533	20.7K	114.4	9.7	4.0	103.9	327.5
MP3D (scans) [7]	90	50,851	1,658	29.6K	328.4	12.8	2.8	95.5	565.0
Gibson tiny (scans) [41]	35	2,397	35	4.9K	139.4	8.8	4.2	15.5	68.5
ReplicaCAD [35]	90	92	39	4.5K	49.8	7.2	4.8	14.3	25.5
iTHOR [23]	120	3,748	112	2.0K	17.1	2.4	9.6	28.2	46.2
RoboTHOR [12]	75	652	47	1.9K	25.9	8.1	5.8	28.8	38.4
ProcTHOR [14]	12K	1,547	95	808.4K	67.4	10.0	5.2	40.5	74.7
HSSD-200 (ours)	211	18,656	466	53.2K	252.2	13.7	5.9	61.5	329.7

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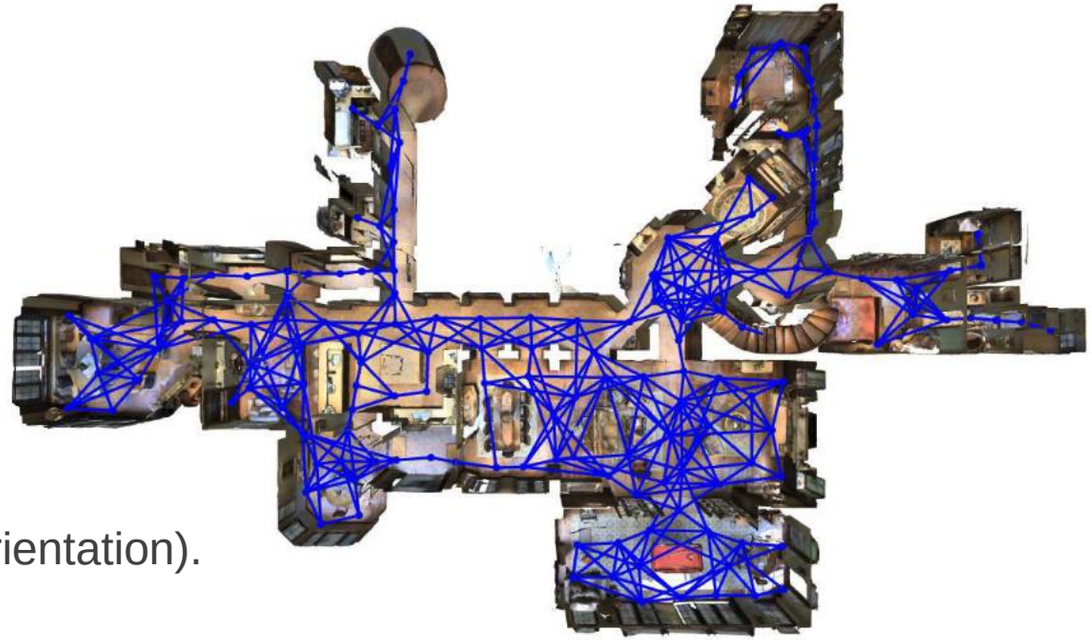
MP3D Simulator



Core Features:

•Discretize scenes into a topological graph.

•Agent navigates only through graph nodes.

Simulation Observation Space:

•12 RGB-D images (30° FOV each).

•Current position in space.

•Nearby navigable points (sorted by angle relative to current orientation).

- First point: current position.

Simulation Action Space:

•Select a nearby navigable point (can stay at current point).

•Choose target rotation angle.

VLN task require the agent to make STOP action in the goal position

Model Action Space:

•Low level action: Move forward, turn left, turn right, look up, look down, stop.

•Navigable point / direction

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R2R benchmark

Task Execution Process：

- **Initialization:**
  - Receive a language instruction L of length T

$$\bar{x} = \langle x_1, x_2, \dots, x_L \rangle$$

  - Start at initial position

$$s_0 = \langle v_0, \psi_0, \theta_0 \rangle \text{ (3D position: xyz, rotation, elevation)}$$

- **First Action:**
  - Capture an RGB image s_0 and generate the first action a_0.

- **Action Execution:**
  - Each action a_t moves the agent to a new position:

$$s_{t+1} = \langle v_{t+1}, \psi_{t+1}, \theta_{t+1} \rangle$$

  - Obtain new observation $s_(t+1)$.

- **Action Sequence:**
  - Execute a series of actions:

$$\langle s_0, a_0, s_1, a_1, \dots, s_T, a_T \rangle$$

- **Success Condition:**
  - Perform "Stop" action within 3m of the target point.

R2R 案例

Pass the pool and go indoors using the double glass doors. Pass the large table with chairs and turn left and wait by the wine bottles that have grapes by them.

Walk straight through the room and exit out the door on the left. Keep going past the large table and turn left. Walk down the hallway and stop when you reach the 2 entry ways. One in front of you and one to your right. The bar area is to your left.

Enter house through double doors, continue straight across dining room, turn left into bar and stop on the circle on the ground.

穿过泳池并使用双层玻璃门进入室内。经过带椅子的大桌子，然后左转，在有葡萄图案的酒瓶旁边等待。直走穿过房间，从左侧的门出去。继续经过大桌子，然后左转。沿着走廊走，当你到达两个入口时停下，一个在你面前，一个在你的右边。酒吧区在你的左边。通过双层门进入房子，直走穿过餐厅，左转进入酒吧，并停在地面上的圆圈处。

Standing in front of the family picture, turn left and walk straight through the bathroom past the tub and mirrors. Go through the doorway and stop when the door to the bathroom is on your right and the door to the closet is to your left.

Walk with the family photo on your right. Continue straight into the bathroom. Walk past the bathtub. Stop in the hall between the bathroom and toilet doorways.

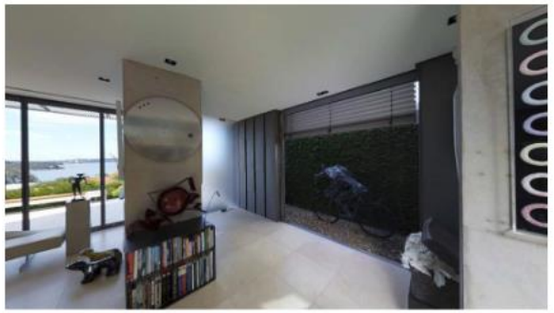
Walk straight passed bathtub and stop with closet on the left and toilet on the right.

站在家庭照片前，向左转，直走穿过浴室，经过浴缸和镜子。穿过门口，当浴室门在你的右边，而衣橱门在你的左边时停下。行走时，让家庭照片保持在你的右侧。继续直走进入浴室。走过浴缸。停在浴室和厕所门之间的走廊上。直走经过浴缸，停在衣橱在左边、厕所在右边的位置。

Exit the office then turn left and then turn left in the hallway and head down the hallway until you get to a door on your left and go into office 359 then stop.

Go out of the room and take a left. Go into the first room on your left.

Leave the office and take a left. Take the next left at the hallway. Walk down the hall and enter the first office on the left. Stop next to the door to office 359.



Go up the stairs and turn right. Go past the bathroom and stop next to the bed.

Walk all the way up the stairs, and immediately turn right. Pass the bathroom on the left, and enter the bedroom that is right there, and stop there.

Walk up the stairs turn right at the top and walk through the doorway continue straight and stop inside the bedroom.

R2R Dataset

Where Do Language Instructions Come From?

1.Instruction Collection:

1. A path is predefined, and humans describe it in detail.
2. In R2R, paths are relatively short, with detailed descriptions.

2.Dataset Statistics:

1. Total of 21,567 language instructions collected.
2. Average instruction length: 29 words.
3. Multiple language instructions may describe the same path.

R2R Dataset Format:

1. Each episode is a pair:
   ⟨Instruction, Image and Action Sequence⟩.
2. Training can be performed offline.

02

Simulators

Summary of simulators

Simulator	Datasets	Observation	Customizable
VizDoom[a]	–	Virtual	✓
House3D	SUNCG	Synthetic	
AI2-THOR[a]	–	Synthetic	✓
Gibson	2D-3D-S, Matterport3D, Gibson	Photorealistic	
iGibson	iGibson	Photorealistic	✓
Matterport3D	Matterport3D	Photorealistic	
Habitat	Gibson, Matterport3D, Replica	Photorealistic	✓

[a]Means that the simulator has a built-in dataset

	Rendering		Physics		Scene	Speed
	Library	Supports	Library	Supports	Complexity	(steps/sec)
Habitat [3]	Magnum	3D scans	none	continuous navigation (navmesh)	building-scale	3,000
AI2-THOR [6]	Unity	Unity	Unity	rigid dynamics, animated interactions	room-scale	30 - 60
ManipulaTHOR [33]	Unity	Unity	Unity	AI2-THOR + manipulation	room-scale	30 - 40
ThreeDWorld [7]	Unity	Unity	Unity (PhysX) + FLEX	rigid + particle dynamics	room/house-scale	5 - 168
SAPIEN [34]	OpenGL/OptiX	configurable	PhysX	rigid/articulated dynamics	object-level	200 - 400†
RLBench [35]	CoppeliaSim (OpenGL)	Gouraud shading	CoppeliaSim (Bullet/ODE)	rigid/articulated dynamics	table-top	1 - 60†
iGibson [36]	PyRender	PBR shading	PyBullet	rigid/articulated dynamics	house-scale	100
Habitat 2.0 (H2.0)	Magnum	3D scans + PBR shading	Bullet	rigid/articulated dynamics + navmesh	house-scale	1,200

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Habitat

1.What is Habitat?

1. A simulation platform for training embodied AI agents.
2. Designed for tasks like navigation, interaction, and scene understanding.
3. Supports both real-world and synthetic 3D environments.

2.Key Features of Habitat:

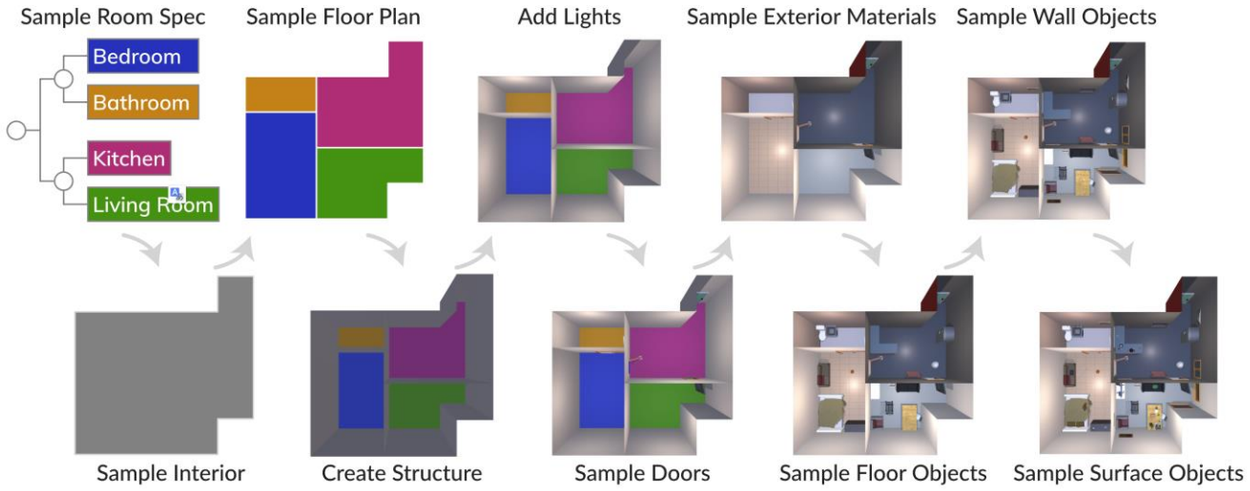
1. **High-performance rendering:** Fast and scalable simulations.
2. **Support for multiple datasets:**
    1. Real-world: Matterport3D (MP3D), HM3D.
    2. Synthetic: ProcTHOR, HSSD.
3. **Sensor simulation:** RGB, depth, semantic segmentation, etc.
4. **Task support:** Object Navigation, VLN, Mobile Manipulation
5. Supports Continuous Navigation Space

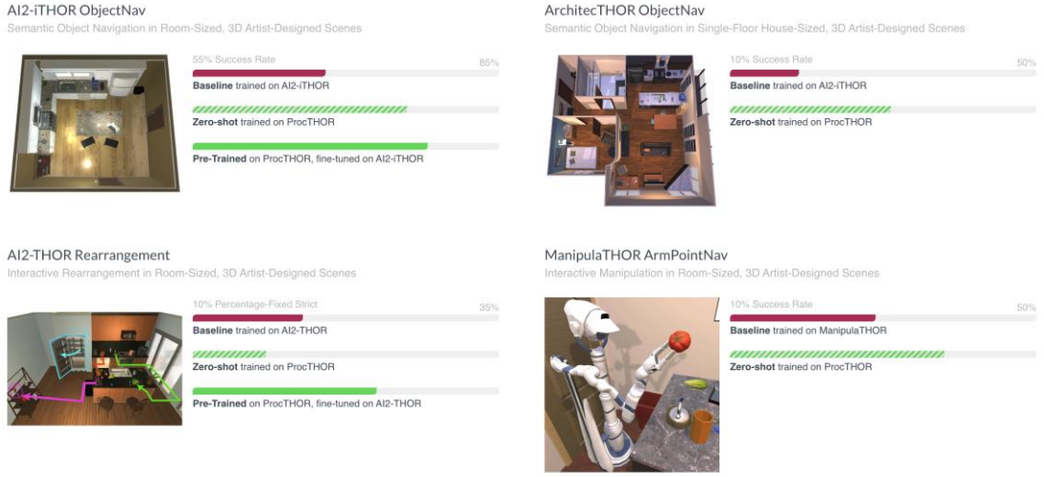
3.Core Components of Habitat:

1. **Habitat Simulator:** The engine for simulating environments and agents.
2. **Habitat API:** Tools for defining tasks, agents, and sensors.
3. **Habitat Lab:** A library for training and evaluating agents.

AI-THOR

1. What is AI-THOR
   - Another embodied AI simulation platform in interactive 3D environments.
   - Built on the Unity engine, providing high-quality visuals and physics.
   - Quite like Habitat in terms of task
   - The related community is operated by Allenai
2. ProcTHOR
   - A large scale procedurally synthetic scene dataset





Isaac Lab

1. What is Isaac Lab/Gym/Sim?

- A suite of tools developed by **NVIDIA** for **robotics simulation and reinforcement learning (RL)**.
- Built on the **NVIDIA Omniverse** platform, leveraging high-performance GPU acceleration.
- Designed to train and evaluate **robotic agents** in realistic, physics-based environments.

2. Key Components

A. Isaac Sim: A high-fidelity **robotics simulator** for creating and testing robotic systems.

B. Isaac Gym: A **reinforcement learning framework** for training robotic agents. #Not longer supported

C. Isaac Lab: Built on Isaac Sim, A lightweight, **task-focused simulation environment** for robotics research.

3. Why Use Isaac Lab/Gym/Sim?

- **High Performance:** GPU-accelerated simulations for faster training.
- **Realism:** High-fidelity physics and rendering for accurate robotic training.
- **Scalability:** Run thousands of environments in parallel for efficient RL.
- **Integration:** Seamless integration with ROS, Omniverse, and machine learning frameworks.

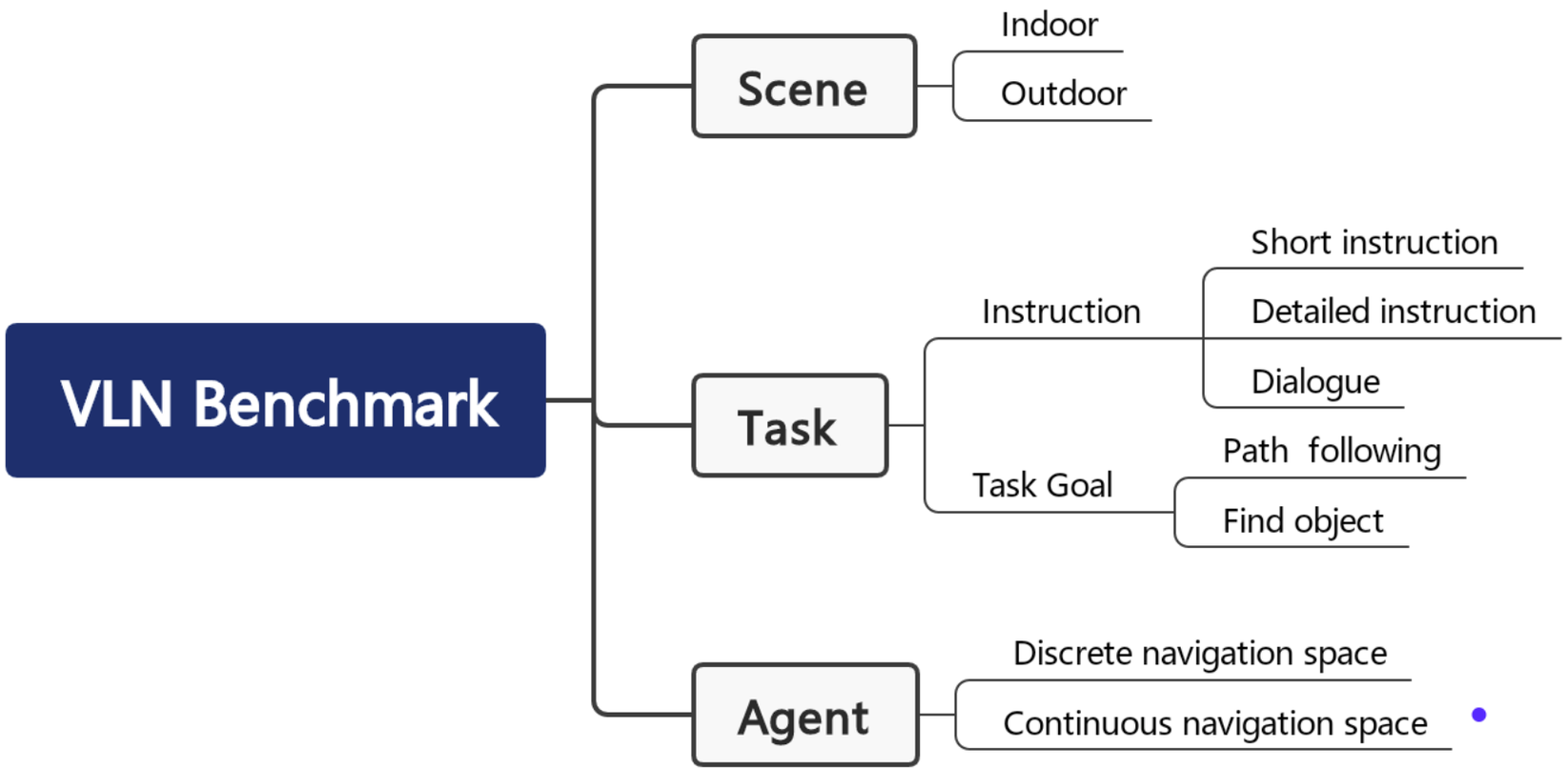
Comparison Different Simulators

- **AI-THOR:** Focused on interactive tasks in fixed environments; less scalable.
- **Habitat:** Optimized for navigation and exploration; lacks high-fidelity physics.
- **Isaac Suite:** Combines **high-fidelity physics**, **scalability**, and **GPU acceleration**, making it ideal for **robotics research**

03

Benchmarks

Structure of VLN benchmark



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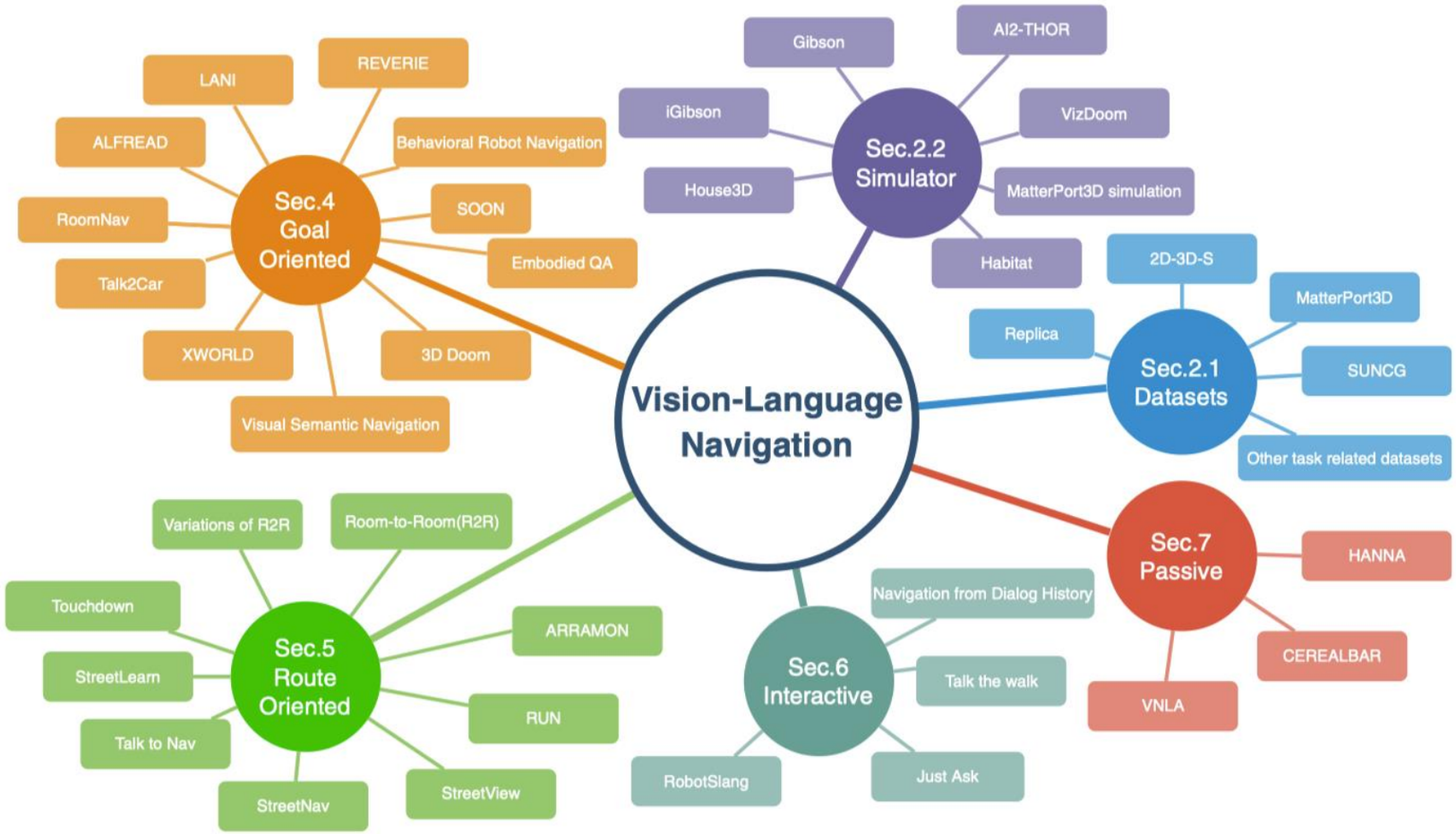
Overview of Benchmarks



(a)

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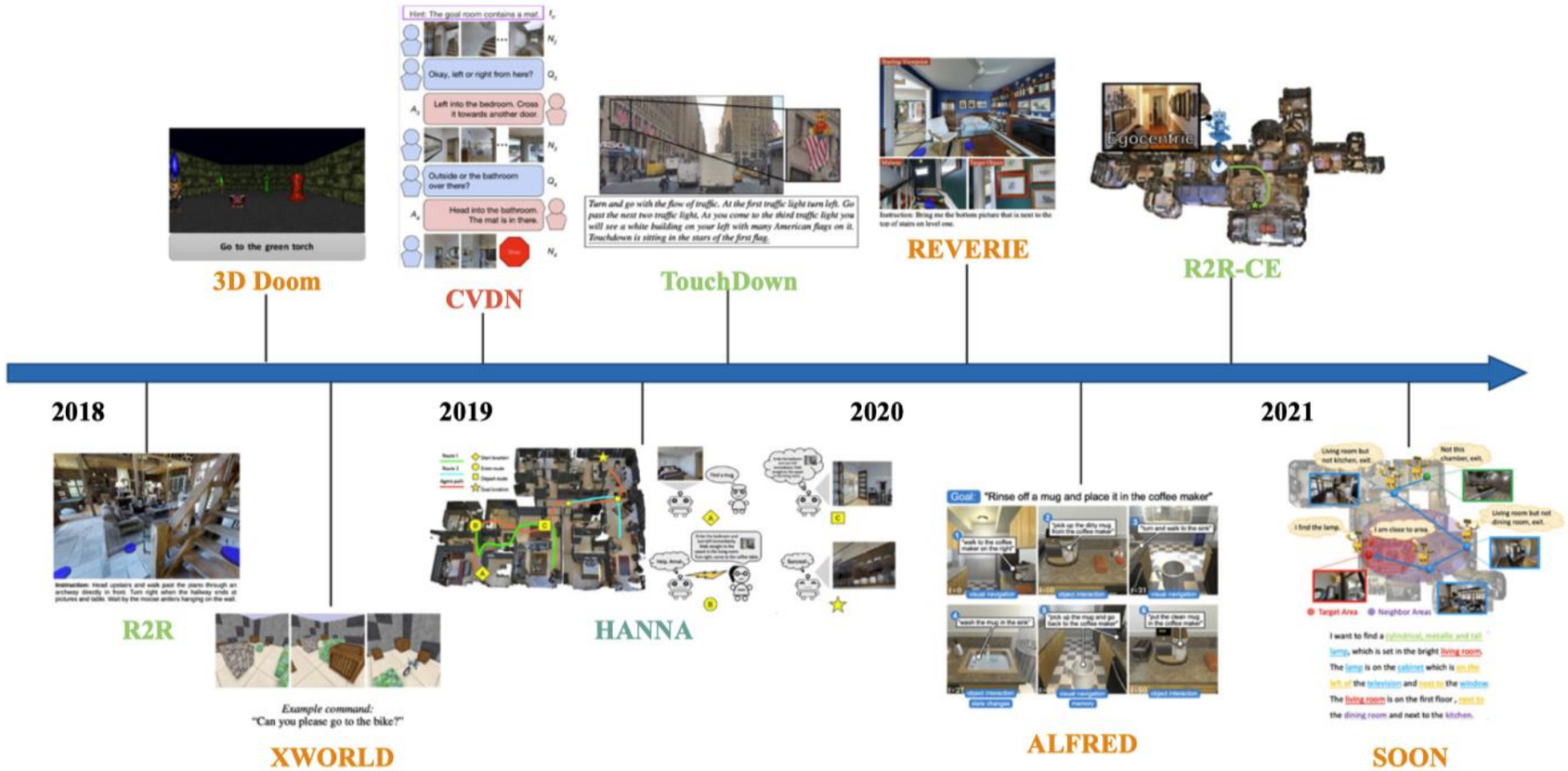
Timeline of Benchmarks



(b)

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Summary of Benchmarks

Task type		Task name	Simulator	Scene type	Action type
Single-Turn	Goal-Oriented	LANI [23]	–		–
		ALFRED [1]	AI2-THOR [16]		M
		Talk to Car [24]	–	Outdoor	M
		XWORLD [25]	–		Q
		3D Doom [26]	VizDoom [17]		–
		Visual Semantic Navigation [27]	AI2-THOR [16]		–
		SOON [28]	Matterport3D [3]		L
		EQA [29]	House3D [21]		Q
		RoomNav [21]	House3D		–
		REVERIE [4]	Matterport3D		L
		Behavioral Robot Navigation [30]	–		–
		Navigation Task Based on SUNCG [31]	–		–
	Route-Oriented	Room-to-Room [3]	Matterport3D		–
		Room-for-Room [32]	Matterport3D		–
		Room-Cross-Room [33]	Matterport3D		–
		R6R, R8R [34]	Matterport3D		–
		Room-to-Room-CE [35]	Matterport3D		–
		Cross-Lingual Room-to-Room [36]	Matterport3D		–
		TouchDown [37]	Google Street View	Outdoor	L
		RUN [38]	–	Outdoor	–
		Street Nav [39]	Google Street View	Outdoor	–
		StreetLearn [40]	Google Street View	Outdoor	–
		ARRAMON [41]	–	Outdoor	M
Multiturn	Passive	CEREALBAR [42]	–		–
		VLNA [43]	Matterport3D		–
		HANNA [44]	Matterport3D		–
	Interactive	CVDN [45]	Matterport3D		–
		Just Ask [46]	Matterport3D		–
		Talk The Walk [47]	–	Outdoor	–
		RobotSlang [48]	Physical		–

The label '–' in the Simulator column means that the corresponding simulator used by this task does not have a special name. In the Scene column, the default type of scene is an indoor scene. In the Compound column, M, Q, and L mean that manipulation, question answering, and location tasks are compounded with navigation tasks, respectively. The label '–' means that only navigation tasks are needed

R*R Serious

1. R4R (Room-for-Room)

•**Goal**: Makes R2R paths **longer and harder**.

•**How**: Combine R2R paths into 3× longer routes.

•**Why**: Tests if agents can follow **twisting paths** without mistakes.

2. RxR (Room-across-Room)

•**Goal**: Add **3 languages** (English, Hindi, Telugu) to R2R.

•**How**: 126,000 instructions with **step-by-step human examples**.

•**Why**: Helps agents understand **different languages and details**.

4. Landmark-RxR

•**Goal**: Add **landmarks** (e.g., sofa, table) to RxR.

•**How**: Instructions include landmarks to guide agents.

•**Why**: Helps agents **find places faster** using objects.

What is VLN-CE?

- **Goal**: Train AI agents to follow language instructions in **realistic 3D environments** (like homes) using **free movement**, not fixed paths.
- **Key Idea**: Remove "cheats" like pre-made maps or teleportation to mimic real-world robot challenges.

Key Features of VLN-CE:

1. **No Fixed Paths**:
    1. Agents move step-by-step (e.g., "turn left 15°," "move forward 0.25m").
    2. No teleporting between pre-defined points.
2. **Longer & Harder Paths**:
    1. Average path: **55 actions** (vs. 4-6 steps in older benchmarks like R2R).
    2. Paths are **3x longer** and more complex.
3. **Dataset**:
    1. Built from **R2R data** but adjusted for continuous movement.
    2. Includes **4,475 paths** with human-written instructions.

Why VLN-CE Matters:

- **Real Robots**: Prepares AI for real-world tasks (no shortcuts like fixed maps).
- **Harder Tests**: Forces agents to *actually navigate* (not just follow a map).

What is REVERIE?

- **Goal**: Train AI agents to **find and point to objects** in 3D homes using **high-level language instructions** (e.g., “Bring me the cushion near the sofa”).

Key Features

1. **Two Tasks in One**:
    1. **Navigate** to the target room.
    2. **Identify** the correct object using a bounding box.
2. **Real-World Complexity**:
    1. Objects are **not visible** at the start.
    2. Instructions are **short and practical** (no step-by-step directions).
3. **Dataset**:
    1. **21,702 instructions** in 86 real 3D homes.
    2. **4,140 objects** (489 categories like chairs, lamps, etc.).
    3. Split into **train**, **validation** (seen/unseen), and **hidden test**.

Success Rules

- **Correct Object**: Select the right bounding box.
- **Within 3 Meters**: Agent must be near the target.
- **One Guess Only**: Agent can’t try multiple times.

Evaluation Metrics

1. **Remote Grounding Success (RGS)**:
    1. Main metric: % of tasks where the agent finds and correctly IDs the object.
2. **Navigation Metrics**:
    1. **Success Rate (SR)**: % of tasks reaching the goal.
    2. **SPL**: Success weighted by path efficiency.
    3. **Path Length (PL)**: Average distance traveled.

Why REVERIE is Hard

- **No Step-by-Step Help**: Instructions are vague (e.g., “Find the lamp near the bed”).
- **Realistic**: Mimics how humans ask robots for help.
- **Unseen Homes**: Tested in new environments to check if agents generalize.

Difference from Other Benchmarks

- **R2R/VLN**: Focus on detailed navigation instructions.
- **REVERIE**: Combines **navigation + object finding** with high-level commands.

Other Becnmarks

- **TOUCHDOWN**:
  - Navigation task in city streets using real images from New York City.
  - TOUCHDOWN happens outdoors, making it more complex due to varied visuals.
- **SOON:**
  - Agents must find a target object in a house using only a description, with no given route.
  - SOON requires agents to plan their own routes and explore.
- **VNLA:**
  - Agents navigate indoor environments to find objects, getting instructions along the way.
  - VNLA provides ongoing guidance during navigation.
- **CVDN:**
  - Agents use past conversations with a human guide to navigate to a goal.
  - CVDN involves dialogues over multiple turns to assist navigation.
- **RoomNav:**
  - Agents are told to go to a type of room, like "Go to the kitchen," and must find it.
  - RoomNav requires understanding room types without guided paths.

Other Becnmarks

- **Just Ask:**
  - Agents can ask questions to a human guide when they are unsure.
- **Talk the Walk:**
  - A task where a tourist and a guide talk to navigate a virtual city and reach a destination.
  - it uses dialogues between two agents and takes place in outdoor city environments.
- **EQA:**
  - Agents explore environments to find answers to questions.
  - EQA emphasizes exploration and reasoning to answer questions.
- **LANI:**
  - Agents follow instructions to reach goals in simple, synthetic environments.
  - LANI focuses on mapping instructions to actions in simpler places.
- **HANNA:**
  - Agents receive simple commands like "find an object" and navigate to locate it.
  - HANNA emphasizes agents deciding their own paths with minimal instructions.

Future Direction

- Unified and Realistic Tasks and Platforms
- Add manipulation
- Dynamic Environment.
- Indoors to Outdoors.

Summary

Name	World		Human			VLN Agent			Dataset	
	Domain	Environment	Turn	Format	Gran.	Type	Act. Sp.	Other	Text	Route
LANI/CHAI (2018)	Indoors	CHALET	Single	Multi Instr	A	-	Disc	Mani	H	H
R2R (2018)	Indoors	Matterport3D	Single	Multi Instr	A	Robot	Graph		H	P
R4R (2019)	Indoors	Matterport3D	Single	Multi Instr	A	Robot	Graph		H	P
RxR (2020)	Indoors	Matterport3D	Single	Multi Instr	A	Robot	Graph		H	P
SOON (2021a)	Indoors	Matterport3D	Single	Multi Instr	G	Robot	Graph		H	P
REVERIE (2020b)	Indoors	Matterport3D	Single	Multi Instr	A, G	Robot	Graph	Detect	H	P
VNLA (2019)	Indoors	Matterport3D	Multi	Multi Instr	A, G	Robot	Graph		T	P
HANNA (2019)	Indoors	Matterport3D	Multi	Multi Instr	A, G	Robot	Graph		H	P
CVDN (2020)	Indoors	Matterport3D	Multi	Restricted	A	Robot	Graph		H	H
VLN-CE (2020)	Indoors	Habitat, Matterport3D	Single	Multi Instr	A	Robot	Disc		H	P
Robo-VLN (2021)	Indoors	Habitat, Matterport3D	Single	Multi Instr	A	Robot	Cont		H	P
RobotSlang (2021)	Indoors	Real	Multi	Freeform	A	Robot	Disc		H	P
ALFRED (2020)	Indoors	AI2-THOR	Single	Multi Instr.	A, G	Robot	Disc	Mani	H	P
TEACh (2022)	Indoors	AI2-THOR	Multi	Freeform	A, G	Robot	Disc	Mani	H	H
DialFRED (2022)	Indoors	AI2-THOR	Multi	Restricted	A, G	Robot	Disc	Mani	H, T	P
TouchDown (2019)	Outdoors	Google Street View	Single	Multi Instr	A	-	Graph		H	P
Street Nav (2020)	Outdoors	Google Street View	Multi	Multi Instr	A	-	Disc		T	P
Talk2Nav (2021)	Outdoors	Google Street View	Single	Multi Instr	A, G	-	Disc		H	P
TtW (2018)	Outdoors	Real	Multi	Freeform	A, G	-	Disc		H	H
LCSD (2019)	Outdoors	CARLA	Single	Multi Instr	A	Driving	Disc		H	P
CDNLI (2020)	Outdoors	CARLA	Multi	Multi Instr	A, G	Driving	Cont		H, T	H
SDN (2022)	Outdoors	CARLA	Multi	Freeform	A, G	Driving	Disc, Cont		H	H
AerialVLN (2023b)	Outdoors	AirSim	Single	Multi Instr	A, G	Aerial	Disc		H	H
ANDH (2023a)	Outdoors	xView	Multi	Freeform	A, G	Aerial	Disc		H	H

World in Which Navigation Occurs:

- **Domain (Indoors or Outdoors)**: inside buildings or outside.
- **Environment**: Dataset

Type of Human Interaction Involved:

- **Turns**: Whether interaction is once or multiple exchanges.
- **Format of Communication**: freeform dialogue, set dialogue patterns, or multiple instructions.
- **Language Granularity**: Level of detail in instructions—specific actions or overall goals.

Action Space and Tasks Assigned to the VLN Agent:

- **Agent Types**: The kind of agent—household robot, self-driving car, or drone.
- **Action Space**: How the agent can move—along predefined paths (graph-based), with limited moves (discrete), or freely (continuous).
- **Additional Tasks**: Other tasks like manipulating objects or detecting items.

Methods of Dataset Collection:

- **Text Collection**: How instructions are gathered—written by humans or generated from templates.
- **Route Demonstrations**: How navigation paths are obtained—from human demonstrations or algorithmic planning.

04

Practice